

ZERO TO JOB-READY DATA ENGINEER 2026

M00 - COMPUTER, SETUP & LEARNING FOUNDATIONS

ABSOLUTE-BEGINNER TEACHING BOOK - REBUILD RC2

Who this book is for

Someone who is completely new to technical study and does not want unexplained jargon, hidden steps or “just know this already” assumptions.

You are not expected to know files, paths, ZIPs, VS Code or terminals before this book teaches them.

How this book teaches

Every lesson follows the same rhythm: understand the idea -> watch the written teacher demonstration -> do it with me -> see the expected result -> understand why it worked -> try a similar task alone -> explain it back.

The short checkpoint-style material is kept for revision. This book is the teacher.

How to use this book

One rule that matters

Do not rush to the checkbox. If a step says 'look at this' or 'explain this', actually do it. The goal is not to finish pages; the goal is to make unfamiliar computer actions feel ordinary.

The lesson rhythm

Stage	What it means
1. Understand	Plain-English mental model before technical words.
2. See it	A written teacher demonstration with exact clicks/commands.
3. Do it with me	You repeat the same action on your PC.
4. Expected result	The book tells you what success should look like.
5. Why it worked	We connect the action back to the idea.
6. Your turn	A similar task without the answer copied in front of you.
7. Teach-back	You explain the idea in your own words.
8. Review	Only then use the short checklist/revision material.

Official reference links

Windows File Explorer help: [Microsoft - File Explorer in Windows](#)

VS Code Windows setup: [Visual Studio Code - Installing on Windows](#)

VS Code beginner overview: [Visual Studio Code - Get Started](#)

Important

These links are optional references. You should be able to complete M00 from this book and the bundled practice files. A link must never be the thing that secretly teaches a missing concept.

Contents

[M00-L01 - Files, Folders, Drives and Paths](#)

[M00-L02 - File Extensions and Common Data Files](#)

[M00-L03 - Download, ZIP, Extract, Copy, Move and Rename](#)

[M00-L04 - Installing Software Safely](#)

[M00-L05 - VS Code From Zero](#)

[M00-L06 - Terminal Fundamentals](#)

[M00-L07 - Errors and Beginner Troubleshooting](#)

[M00-L08 - Study Evidence, Attempts and Review Discipline](#)

[M00 Summary - What you should be able to do before M01](#)

M00-L01 - Files, Folders, Drives and Paths

By the end of this lesson

You can look at a file on your computer and answer three questions: What is it? Which folder is it inside? What is its exact address/path?

[Back to Contents](#)

Teacher: Before data engineering, before Python, before SQL, there is one boring-looking skill that prevents hundreds of future problems: knowing exactly where your files are. If I give you a file called orders.csv and you cannot tell me where it lives, every later tool becomes harder.

1. Start with something you already understand: a house address

Imagine someone says, 'Come to Ashish's house.' That is not enough for a delivery person. They need a complete address. A file name works the same way. 'orders.csv' is only the name. A path is the address that tells the computer where that file lives.

Word	Beginner meaning
Drive	A large storage area. On many Windows PCs the main drive is C:.
Folder	A container that keeps files and other folders organized.
File	One saved item: a PDF, picture, spreadsheet, Python program, text note, etc.
Path	The address of a file or folder.

Before we continue: Desktop, Documents and Downloads

These are ordinary folders in your Windows user account. Desktop is the folder whose items appear on your desktop screen. Documents is a sensible long-term place for study/project folders. Downloads is where browsers usually place newly downloaded files. A downloaded file is not 'inside Chrome'; Chrome only placed it into a folder on your drive.

Beginner check

If you download a PDF and close Chrome, the PDF still exists because the file was saved to storage. The browser and the file are separate things.

2. Read a path like an address

```
C:\Users\Student\Documents\DE_COURSE\data\orders.csv
```

Read it from left to right: start on drive C:, enter Users, enter Student, enter Documents, enter DE_COURSE, enter data, and finally you reach the file orders.csv. Each backslash separates one level from the next.

Notice the difference

orders.csv = only the file name.

C:\Users\Student\Documents\DE_COURSE\data\orders.csv = the full location.

A path tells you location, not what the file contains

Two files can have the same name and still be different files if they live in different folders. For example, Downloads\orders.csv and Documents\Project\orders.csv can both exist at the same time. This is why 'I opened orders.csv' is not enough when troubleshooting; we often need the full path.

Two same names, two different files

Downloads\orders.csv

Documents\DE_COURSE_PRACTICE\data\orders.csv

3. Do it with me in File Explorer

1. Press Windows key + E. This opens File Explorer.
2. In the left side, choose Documents. If your Windows opens to Home first, that is normal.
3. Right-click an empty area -> New -> Folder. Name it DE_COURSE_PRACTICE.
4. Open DE_COURSE_PRACTICE. Create another folder named files_lab.

5. Open files_lab. Right-click -> New -> Text Document. Name it location_test.txt.
6. Click once in the File Explorer address bar near the top. Windows will show the folder location.
7. Look at the path and say out loud: “This is the address of the folder that contains location_test.txt.”

Expected result

You should have exactly one DE_COURSE_PRACTICE folder, inside it one files_lab folder, and inside that one location_test.txt file. Closing File Explorer and reopening it should not make the file disappear.

4. Nested folders are just boxes inside boxes

```
DE_COURSE_PRACTICE/
01_MODULE_WORK/
M00/
  data/
  notes/
  evidence/
```

There is nothing magical about a 'project structure'. It is simply an agreed arrangement of folders so that you and your tools know where things belong.

5. Absolute vs relative path - only the idea for now

An absolute path gives the complete address from a fixed starting point, such as C:. A relative path says, 'from where I am now, go here.' You do not need to master this yet. Later, Python, Git and Linux will use it constantly.

Example

Absolute: C:\Users\Student\Documents\DE_COURSE_PRACTICE\files_lab\location_test.txt

Relative (if you are already inside DE_COURSE_PRACTICE): files_lab\location_test.txt

6. Beginner mistakes and why they happen

- You save something in Downloads, then later look in Documents. The file is not gone; you are looking at the wrong address.
- You create project, project2, project_final, project_final2. Later you do not know which is real. Stable names beat “final-final-new”.
- You rely on Recent files. Recent remembers what you opened, not where your project should live.

Quick teach-back

Say these answers in your own words

What is the difference between a file and a folder?

If I tell you only “orders.csv”, what information is missing?

What does a path tell you?

Your turn - do not copy

Independent mini-task

Create another folder inside files_lab named practice. Put a new file called my_address_test.txt inside it. Close File Explorer. Reopen it from Documents and find the file without using Recent. Then write its full path in the file.

M00-L02 - File Extensions and Common Data Files

By the end of this lesson

You can look at .xlsx, .csv, .json, .txt, .sql, .py, .zip and .pdf and explain what kind of file each one is, without confusing “opens in Excel” with “is an Excel workbook”.

[Back to Contents](#)

Teacher: A beginner often sees an icon and thinks the icon tells the truth. It does not. The important clue is the extension - the letters after the last dot in the file name.

1. What is an extension?

```
orders.csv
report.xlsx
pipeline.py
notes.txt
```

In orders.csv, 'orders' is the name you chose and '.csv' is the extension. Windows uses the extension to help decide which apps can open the file. Microsoft also explains that changing the letters at the end does not magically convert the internal file format.

Extension	What it really is	Common app
.xlsx	Excel workbook	Excel or compatible spreadsheet app
.csv	Plain-text table	Excel, text editor, Python, database tools

Extension	What it really is	Common app
.json	Structured text data	VS Code / text editor / Python
.txt	Plain text	Text editor
.sql	SQL script	VS Code / database client
.py	Python source code	VS Code / Python
.zip	Compressed archive	File Explorer / 7-Zip
.pdf	Fixed-layout document	PDF reader / browser

Text files vs richer/binary formats - only the beginner idea

TXT, CSV, JSON, SQL and Python source are readable text formats. If you open them in a plain text editor, you can usually see meaningful characters. XLSX and PDF are more structured formats: opening their raw internal bytes in Notepad is not a useful way to read them. You do not need the word 'binary' deeply yet; just learn that file formats store information differently.

Why this matters later

Python can read a CSV as text-like tabular data. Excel can open that same CSV in cells. But an XLSX workbook contains a richer internal structure. Tools must know which format they are reading.

2. The most important beginner trap: CSV is not XLSX

A CSV often opens in Excel, which makes it look like an Excel file. But CSV is just text arranged in rows and columns. It does not normally carry multiple sheets, formulas, PivotTables or rich workbook formatting. The app you use to open a file and the file's actual format are two different ideas.

Think of it like this

A photograph can be shown inside WhatsApp, Chrome or Photos. That does not make the photograph a 'WhatsApp file'. In the same way, a CSV can open in Excel without becoming an XLSX workbook.

3. Make Windows show the real extensions

1. Open File Explorer.
2. Choose View -> Show -> File name extensions. On some Windows versions the wording can vary slightly.
3. Look at several files. You should now see endings such as .pdf, .jpg, .zip or .txt.
4. Leave extensions visible during this course. It makes technical work safer.

Optional official reference: [Microsoft - Common file name extensions in Windows](#)

Default app is a preference, not the file identity

Windows may remember that you usually open .csv files with Excel or .json files with VS Code. That default-app choice controls what happens when you double-click. It does not change what the file is. You can right-click -> Open with to choose a different suitable app for the same file.

4. Do it with me: open the same CSV two ways

1. Extract the bundled DOWNLOAD_ME_FIRST.zip if you have not already.
2. Open data\sample.csv with Notepad or VS Code first.
3. You should see ordinary text with commas, for example: order_id,customer,amount.
4. Now open the same sample.csv in Excel if Excel is installed.
5. Excel will arrange the comma-separated text into cells. The file is still CSV.

Expected result

Text editor: you can literally see commas separating values.

Excel: the same values appear in rows/columns because Excel knows how to interpret CSV text.

5. Renaming is not converting

WRONG IDEA:

report.csv -> rename to -> report.xlsx

That only changes the label on the container. It does not rebuild the internal format as a real Excel workbook. Proper conversion means opening/importing the data and saving/exporting it in the new format.

6. Common mistakes

- Hiding extensions and accidentally creating report.csv.csv.
- Opening .py or .sql files in Word and adding formatting that can damage code text.
- Assuming a file format based only on the app icon.

Quick teach-back

Say these answers in your own words

What part of report.csv is the extension?

Why can a CSV open in Excel without being an Excel workbook?

Why does renaming .csv to .xlsx not convert the file?

Your turn - do not copy

Independent mini-task

Pick four files from the practice bundle. Before opening them, write what each extension tells you. Then open each with a sensible app and check whether your prediction made sense.

M00-L03 - Download, ZIP, Extract, Copy, Move and Rename

By the end of this lesson

You can download a ZIP, extract it safely, explain copy vs move, and avoid working directly inside a compressed archive.

[Back to Contents](#)

Teacher: A ZIP is not a mysterious technical file. Imagine putting many papers into one sealed travel bag so it is easier to carry. Extraction is unpacking the bag before you work with the contents.

1. ZIP and normal folder are not the same

Thing	What it means
ZIP archive	A compressed container that can hold many files/folders in one downloadable item.
Extracted folder	A normal folder containing the usable working files.

Windows may let you peek inside a ZIP. That preview can look almost like a normal folder, which is exactly why beginners get confused. Unless the instruction says otherwise, extract first and work in the extracted folder.

2. Copy vs move

Action	Original stays in old place?	Use it when
Copy	Yes	You want a second copy while keeping the original.
Move	No - location changes	You are reorganizing the original into a new location.
Rename	Same file, new name	You want a clearer name without changing the file format.

What extraction actually changes

Extraction creates normal copies of the archived files outside the ZIP container. The ZIP is not automatically destroyed. That is why, after extraction, you can see both the original archive and a normal folder. If you edit a file in the extracted folder, the untouched ZIP can still act as a clean starting copy.

Copy, move and extract are different

Copy: make another copy. Move: change location. Extract: unpack files from an archive into normal files/folders.

3. Do it with me: safe download/extract workflow

1. Find DOWNLOAD_ME_FIRST.zip supplied with this rebuilt module.
2. Copy it to Downloads. We are pretending it just arrived from the internet.
3. In Downloads, right-click the ZIP -> Extract All.
4. Choose the normal extraction location offered by Windows and extract.
5. Open the extracted folder. Confirm you can see folders such as data, notes and exercises.
6. Rename the extracted working folder to m00_download_lab.
7. Move m00_download_lab into Documents\DE_COURSE_PRACTICE\01_MODULE_WORK\M00.
8. Keep the original ZIP untouched as your clean source copy for now.

Expected result

The .zip file still exists as a ZIP.

A separate normal folder named m00_download_lab exists inside your course workspace.

You edit files in the extracted folder, not inside the ZIP.

Keyboard actions you will see often

Action	Meaning
Ctrl+C	Copy selected item(s).
Ctrl+V	Paste the copied item(s).
Ctrl+X	Cut: prepare selected item(s) to be moved.
F2 or right-click Rename	Rename the selected item without changing its contents.

Do not use shortcuts blindly

The shortcut is not the skill. The skill is knowing whether you want a second copy, a moved original, or only a new name.

4. Why this habit matters later

Data engineering projects often arrive as repositories, archives, exports and downloaded datasets. If you do not know which copy is the untouched source and which copy is your working version, you can accidentally destroy the original evidence or edit the wrong file.

5. Common mistakes

- Double-clicking the ZIP, editing a file in the preview, then wondering where the saved change went.

- Using Cut/Move when you intended to preserve an original backup.
- Renaming the extension while thinking you changed the real format.
- Downloading the same file five times instead of locating the first download.

Quick teach-back

Say these answers in your own words

What does extracting a ZIP do?

When would you choose Copy instead of Move?

Why should you keep a clean source copy during practice?

Your turn - do not copy

Independent mini-task

Take the bundled messy_downloads folder. Make a copy named messy_downloads_working. Organize only the working copy. Keep the source untouched.

M00-L04 - Installing Software Safely

By the end of this lesson

You can distinguish an installer from an installed program, use an official download source, understand the basic idea of PATH, and verify that a program actually works after installation.

[Back to Contents](#)

Teacher: Installing software is not the same as downloading software. The file you download is usually the installer - a helper whose job is to put the real program in the correct places and register it with Windows.

1. Downloaded installer vs installed application

Thing	Meaning
Installer file	A setup program such as VSCodeUserSetup-x64.exe. You run it to install the application.
Installed application	The actual program available from Start menu / shortcuts / command line after setup finishes.

Simple analogy

A food-delivery box is not the restaurant. The installer delivers and arranges the program; after installation, you use the installed program, not the installer every day.

2. Safe source rule

Whenever possible, use the official product website or official app store. Search-result pages can contain advertisements with large 'Download' buttons that belong to unrelated software.

- Check the website domain before downloading.
- Do not install random download managers, browser helpers or “PC optimizer” offers just because they appear near the real download button.
- Read installer screens instead of clicking Next automatically.
- Install only the tool the current module actually needs.

What Windows permission prompts mean

Some installers ask for administrator permission through a Windows User Account Control (UAC) prompt. That prompt means the installer wants permission to make system-level changes. Do not click Yes automatically just because a popup appears. First confirm that you intentionally launched the trusted installer you expected.

A safe pause

Ask: Did I start this installer? Is the product name what I expected? Did I get it from the official source? If the answer is unclear, cancel and check before continuing.

3. VS Code example

Official Windows installation page: code.visualstudio.com/docs/setup/windows

The official VS Code documentation recommends User Setup for most Windows users. It normally installs for your Windows account and does not require administrator permissions. If VS Code is already installed and working, do not reinstall it just because this lesson exists.

32-bit / 64-bit and versions - what a beginner needs

Modern Windows PCs are usually 64-bit, and many download pages offer multiple installers. Follow the course's exact platform instruction instead of guessing. A version number such as 1.95 or 3.13 identifies a release of the software. Later, version mismatches matter because a tutorial or library can behave differently across releases.

Do not chase the newest version blindly

When a later module specifies an exact or supported version range, use that instruction. 'Newest' is not always the same as 'compatible with this lab'.

4. What is PATH? Beginner mental model

Later you will type commands such as python, git and code. The terminal needs a way to find the program that owns each command. PATH is essentially a list of folders the terminal is allowed to search automatically.

Imagine a contact list

If you say 'call Ravi', your phone searches the saved contacts to find Ravi's number. When you type a command, the terminal searches PATH to find the matching program. A program can be installed but still not be found from the terminal if the command location is not available in PATH or the terminal has not been restarted.

5. Verification: how do I know installation worked?

1. Open the program from the Start menu.
2. Confirm the main window opens without immediately crashing.
3. If the course asks for a version, open Help -> About or the program's version screen.
4. If the tool has a command-line check, run it only when the course tells you. Do not guess random commands.
5. Record the version/result in your completion evidence when requested.

Do not install everything today

M00 teaches the idea of installation. PostgreSQL, Docker, Spark, Kafka and other engineering tools belong to later modules. Installing them early only creates extra failure points.

Quick teach-back

Say these answers in your own words

What is the difference between an installer and the installed application?

Why is an official download source safer?

What does PATH do at a basic level?

Your turn - do not copy

Independent mini-task

Choose one application you already have, such as VS Code. Find where you launch the installed app. Then

locate (if still present) the original installer in Downloads. Explain why these are not the same thing.

M00-L05 - VS Code From Zero

By the end of this lesson

You can open the correct project folder in VS Code, understand Explorer/Editor/Tabs/Terminal, create and save a file in the intended folder, and reopen the same workspace later.

[Back to Contents](#)

Teacher: VS Code is a workspace for technical files. It is not Python, not SQL, not a database and not Git. Think of it as a smart desk where project files, an editor and a terminal can sit together.

1. Four areas you need today

Area	What it is
Explorer	Left panel that shows files/folders in the project you opened.
Editor	Large middle area where the selected file is displayed and edited.
Tabs	Labels near the top that let you switch between open files.
Terminal	Panel where you type commands from the project environment.

A project is usually a folder, not one magic file

Technical work normally uses several related files together: data, code, notes, configuration and evidence. VS Code becomes much more useful when it can see the whole project folder. The Explorer panel is simply showing that folder tree inside the editor.

If you feel lost in VS Code

Look at the Explorer root name first. Ask: 'Which folder did I open?' If the root is wrong, close the folder and use File -> Open Folder again.

2. The project-root idea

Suppose your project looks like this:

```
supplier_import/
data/
supplier.csv
```

```
src/  
  clean_supplier.py  
notes/  
  requirements.txt
```

The project root is `supplier_import` - the top folder that contains the whole project. If you open only `clean_supplier.py`, VS Code can show that file, but it does not automatically give you the same clear project context.

Rule

For course projects, prefer File -> Open Folder -> choose the project root.

3. Do it with me

1. Open VS Code.
2. Choose File -> Open Folder.
3. Choose Documents\DE_COURSE_PRACTICE\01_MODULE_WORK\M00\m00_download_lab.
4. If Windows asks whether you trust the authors of the files, read the message. For our bundled practice files, you can trust the folder.
5. Look at the Explorer panel on the left. Find notes and data.
6. Open notes\instructions.txt by clicking it once.
7. In the Explorer, create a new file at the project root named evidence.txt.
8. Type: VS Code workspace opened successfully.
9. Press Ctrl+S to save.
10. Close VS Code. Open it again and reopen the same folder. Confirm evidence.txt is still there.

Expected result

Explorer shows the whole `m00_download_lab` project tree.

`evidence.txt` appears beside the top-level folders, not accidentally inside data or notes.

Closing and reopening does not erase the saved file.

What happens when you click a file in Explorer

VS Code reads the file from your storage drive and shows its contents in an editor tab. When you change text, the in-memory editor becomes different from the saved file until you save. Ctrl+S writes the new contents back to the file on disk. That is why an unsaved indicator matters.

Simple mental model

Explorer = where the file lives. Editor = what is inside the selected file. Save = write your edited contents back to storage.

4. Saved file vs unsaved tab

An editor can hold text that has not yet been saved to disk. A tab named Untitled is a warning that you may be editing something that does not yet have a real file location. Saving gives the text a name and a place on your storage drive.

5. Common mistakes

- Opening one random file instead of the whole project root.
- Creating evidence.txt inside the wrong subfolder because you did not notice which folder was selected.
- Typing into Untitled and closing before saving.
- Installing many VS Code extensions before you know why you need them.

Quick teach-back

Say these answers in your own words

What is VS Code for?

What is the project root in a folder tree?

What is the difference between the Explorer and the Editor?

Your turn - do not copy

Independent mini-task

Open the bundled supplier_import_assets folder as a project root. Create evidence/hello.txt at the correct level. Close VS Code and reopen the project. Prove the file stayed where you intended.

M00-L06 - Terminal Fundamentals

By the end of this lesson

You can explain prompt/current directory/command/argument, then safely use pwd, ls, cd, cd .. and mkdir without being afraid of the terminal.

Teacher: A terminal is just another way to tell the computer what to do. File Explorer uses clicks. A terminal uses typed instructions. The scary-looking text becomes much easier once you understand one idea: every command runs from a current folder.

1. Four words before any command

Word	Plain meaning
Prompt	The place where the terminal is waiting for you to type.
Current directory	The folder the terminal is currently working inside.
Command	The action word, such as <code>cd</code> or <code>mkdir</code> .
Argument	Extra information given to the command, such as the folder name.

Terminal vs shell - only the idea

The terminal is the window/panel where you type. The shell is the command interpreter running inside it, such as PowerShell, Command Prompt, Git Bash or later WSL. Different shells can format prompts differently and sometimes use different commands, but the core idea is the same: you type an instruction, the shell interprets it, and the terminal shows the result.

Do not panic when your prompt looks different

Your prompt may not match a screenshot character-for-character. Focus on the command, the current directory and the output.

2. Read a command like a sentence

```
mkdir evidence
```

Command = `mkdir` (make directory/folder). Argument = `evidence` (the name of the folder to make). The computer reads it like: 'make a folder named evidence here'.

```
cd data
```

Command = `cd` (change directory). Argument = `data`. Meaning: 'move my terminal's current location into the data folder'.

3. The current directory is the hidden cause of many errors

If you are standing in the wrong room, saying 'open the cupboard' may refer to the wrong cupboard. A terminal command can fail for the same reason: you are asking for a file relative to the wrong current directory.

Paths with spaces

A folder name such as My Project contains a space. Some shells need you to put such a path in quotes so the whole thing is treated as one argument, for example `cd 'My Project'` or `cd "My Project"` depending on the shell. We avoid complicated paths in early labs, but you should know why quotes sometimes appear.

Why we use simple project names

Names such as `supplier_import` or `DE_COURSE_PRACTICE` reduce quoting mistakes while you are learning. Later you will handle more complex paths confidently.

4. Do it with me in VS Code Terminal

1. Open `m00_download_lab` as a VS Code folder.
2. Open Terminal -> New Terminal.
3. Look at the prompt. Do not type yet. Notice that it usually shows some clue about your current location.
4. Type `pwd` and press Enter. In PowerShell and Git Bash this prints the current path.
5. Type `ls` and press Enter. You should see the project contents.
6. Type `cd data` and press Enter.
7. Type `pwd` again. The path should now end with `data`.
8. Type `ls`. You should see `sample.csv`.
9. Type `cd ..` and press Enter. Two dots mean “parent folder - one level up”.
10. Type `mkdir terminal_created` and press Enter.
11. Look at VS Code Explorer. A new `terminal_created` folder should appear in the project root.

Expected result

After `cd data`, `pwd` ends in `/data` or `\data` depending on shell formatting.

After `cd ..`, you return to the project root.

`terminal_created` appears in the Explorer without you clicking New Folder.

5. Commands to know - not memorize blindly

Command	Question it answers	Meaning
<code>pwd</code>	Where am I?	Print current working directory
<code>ls</code>	What is here?	List files/folders
<code>cd folder_name</code>	Go inside this folder	Change directory
<code>cd ..</code>	Go one level up	Move to parent directory
<code>mkdir name</code>	Create a folder here	Make directory

Safety rule

M00 intentionally avoids destructive terminal commands. Do not experiment with delete/remove commands just because you see them online. First learn location and navigation.

6. Common mistakes

- Typing a file path alone and expecting it to act like a command.
- Running the right command in the wrong directory.
- Forgetting that spaces in some paths may require quotes.
- Copying a terminal command from the internet without understanding what it changes.

Quick teach-back**Say these answers in your own words**

What is the current directory?

In “mkdir evidence”, which word is the command and which is the argument?

What does `cd ..` do?

Your turn - do not copy**Independent mini-task**

From the `supplier_import` project root, use only the terminal to enter data, list the files, return to root, create archive, enter archive and create a folder named 2026. Then use `pwd` at each level and explain how the path changes.

M00-L07 - Errors and Beginner Troubleshooting

By the end of this lesson

You can respond to a simple technical error with a calm, repeatable process instead of changing random settings or reinstalling everything.

[Back to Contents](#)

Teacher: An error message is not the computer insulting you. It is evidence. Your job is not to feel good about the red text; your job is to extract clues from it.

1. Why random fixing is dangerous

Suppose a file is in the wrong folder. If you reinstall VS Code, change Windows permissions, rename three folders and reboot, one of those actions might accidentally make the symptom disappear - but you will not know what the real cause was. Next time you are stuck again.

Symptom, cause and fix are three different things

Word	Meaning	Example
Symptom	What you observe	File not found.
Cause	Why it actually happened	The file is in the wrong folder.
Fix	The change that addresses the cause	Put/use the file at the expected path.

A restart may remove a symptom without fixing the cause. Good troubleshooting tries to identify the cause, not merely make the message disappear once.

2. The troubleshooting loop

1. Reproduce the problem once. Do not click randomly ten times.
2. Read the complete error or symptom.
3. Identify the file, path, command or program mentioned.
4. Form one simple hypothesis: "Maybe I am in the wrong folder."
5. Change one thing only.
6. Retry the same action.
7. If it works, write down what caused it. If it does not, rethink the hypothesis instead of changing five more things.

The key sentence

One hypothesis -> one change -> one retry.

3. Worked example: file not found

Instruction expects:
data/orders.csv

Actual file is here:
data/archive/orders.csv

The important clue is the location. The file exists, but not where the instruction expects it. Reinstalling your editor cannot fix a wrong path. You either move/copy the intended file into the expected location or update the path reference - depending on the task.

Symptom	Inspect first	Beginner question
File not found	Current folder + exact path	Is the file where the command expects it?
Command not recognized	Installation + PATH + terminal restart	Can the shell find the program?
Permission denied	Location + permissions	Are you allowed to change this file/folder?
Cannot open file	File format + app + corruption	Are you using a suitable application?
Unexpected file name	Visible extension	Is it report.csv.csv or a hidden extension issue?

4. Do it with me: controlled broken setup

1. Open exercises\broken_setup\TASK.txt from the practice bundle.
2. Read it completely before moving anything.
3. Make exactly one attempt.
4. Use the symptom to locate the intended input file.
5. Repair only the location problem described by the task.
6. Create repair_note.txt and write one sentence: what was wrong, what you changed, why that fixed it.
7. Keep the original broken_setup_source untouched so you can compare before/after.

A good repair note example

The task expected orders.csv inside data, but the file was inside data/archive. I copied the working file to data so the requested relative path could find it; I kept the original source copy unchanged.

A second worked example: command not recognized

```
python --version
'python' is not recognized ...
```

This symptom does not automatically mean Python is broken. Possibilities include: Python is not installed, the terminal was opened before installation and needs restarting, or the command is not available through PATH. The correct next step is to inspect installation/version instructions - not to change unrelated files.

5. Common mistakes

- Searching the internet before reading the entire local error message.
- Changing many settings at once.
- Reinstalling software to fix a path problem.
- Deleting the broken original so you cannot prove what was wrong.

Quick teach-back

Say these answers in your own words

Why should you change one thing at a time?

If you see “file not found”, what should you inspect before reinstalling software?

What makes a repair note useful?

Your turn - do not copy

Independent mini-task

Open exercises/path_problem/TASK.txt. Save an attempt note first. Fix the path issue using one hypothesis at a time. Then write a two-sentence explanation of the cause and repair.

M00-L08 - Study Evidence, Attempts and Review Discipline

By the end of this lesson

You can separate teacher examples, your independent attempts, review material, retests and evidence so “completed” means real skill rather than copied work.

[Back to Contents](#)

Teacher: If you see the answer before you attempt the task, your brain can confuse recognition with ability. 'Yes, that makes sense when I see it' is not the same as 'I can produce it myself.' This course protects the difference.

1. Five states of learning

Folder/state	Purpose
LEARNING	Teacher explanations, guided examples, follow-with-me work.
ATTEMPTS	Your independent work before opening the answer/review.
REVIEW	Explanations and solutions used after an attempt.
RETESTS	A fresh, changed task after repair.
EVIDENCE	Screenshots, logs, outputs or files that prove what happened.

Recognition vs recall

Recognition means an answer looks familiar when you see it. Recall means you can produce the idea or action without the answer in front of you. Most job tasks require recall plus reasoning. That is why this course separates the guided example from the independent attempt.

Example

Seeing `cd ..` and thinking 'yes, I remember that' is recognition. Opening a terminal later and deciding on your own that `cd ..` is the command you need is recall.

2. Why keep a wrong attempt?

Because the wrong attempt contains information about what you actually knew at that moment. If you overwrite it after seeing the answer, the learning record becomes fake. A wrong attempt can be more useful than a copied correct answer because it tells us what to repair.

Normal flow

Attempt -> Save -> Check -> Review -> Repair -> Fresh retest

3. Do it with me: create your evidence structure

1. Inside `DE_COURSE_PRACTICE` create `02_ATTEMPTS`, `03_REVIEW`, `04_RETESTS` and `05_EVIDENCE` if your workspace does not already use equivalent folders.
2. Inside `02_ATTEMPTS` create `m00_test_attempt.txt`.
3. Type: wrong first attempt kept on purpose.
4. Save it. Do not “correct” that original file later.
5. Create a separate retest file inside `04_RETESTS` named `m00_test_retest.txt`.
6. Write a corrected sentence in the retest file.
7. In `05_EVIDENCE` create `m00_evidence_note.txt` explaining which file is the original attempt and which is the retest.

Expected result

You have two different files showing two different moments in learning. The first attempt is preserved, and the later retest is separate.

Naming evidence so future-you understands it

Evidence names should tell you what happened without opening ten files. Compare final.txt with M00_G03_terminal_navigation_pass.txt. The second name is longer, but six weeks later it still tells you what the file proves.

Example name	What it tells you
M00_P03_attempt1.txt	First independent attempt for practice task P03.
M00_P03_retest1.txt	Fresh proof after review/repair.
M00_G03_terminal_evidence.txt	Evidence from guided terminal lab G03.

4. Mastered, skipped and reviewed are different

Status	Meaning	Interpretation
Mastered	You can reproduce the skill without copying.	Good evidence
Reviewed	You looked at explanation/solution after trying.	Not automatically mastery
Skipped	You intentionally moved past it.	Not mastery

5. Why this matters in job preparation

In an interview or job, the answer sheet disappears. You need the ability to reconstruct the work, explain your reasoning and notice when the result is wrong. Preserving attempts trains exactly that independence.

Quick teach-back

Say these answers in your own words

Why do we save the first attempt before review?

What is the difference between review and retest?

Why is “skipped” not the same as “mastered”?

Your turn - do not copy

Independent mini-task

Create attempts/P05_attempt1.txt with an intentionally wrong sentence. After saving it, create retests/P05_retest1.txt with a corrected version. Do not edit attempt1. In evidence/P05_evidence.txt, explain why both files exist.

M00 Summary - what you should be able to do now

Do not move to M01 just because you reached this page

Move on when these actions feel understandable and reproducible. You do not need perfect speed or memorized commands.

You can now...

- Create a stable course workspace and find it again after closing File Explorer.
- Explain file, folder, drive and path in plain English.
- Recognize common data/file extensions and explain why CSV is not XLSX.
- Extract ZIPs and distinguish source copy from working copy.
- Install software from official sources only when needed and verify that it opens.
- Open the correct project root in VS Code and save files in the intended folder.
- Use pwd, ls, cd, cd .. and mkdir while understanding the current directory.
- Troubleshoot a simple path/location problem using one hypothesis at a time.
- Preserve attempts before review and create separate retest evidence.

Before independent practice**Ready check**

1. I can find DE_COURSE_PRACTICE without using Recent files.
2. I extracted DOWNLOAD_ME_FIRST.zip and can identify the working folder.
3. I opened a project root in VS Code.
4. I created at least one folder from the terminal and can explain where it appeared.
5. I preserved a first attempt and a separate retest.

[Back to Contents](#)